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END TERM EXAMINATION
SECOND SEMESTER [B.TECH] MAY-JUNE 2025

Paper Code: BS-106

Subject: Applied Physics-II

Time: 3 Hours

Maximum Marks: 60

Note: Attempt five questions in all including Q.No. 1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q. No.	Question	Marks
Q1	Answer any four of the following questions:	(5×4=20)
(a)	What are Matter waves? How Davisson and Germer demonstrated their existence in the laboratory?	(5)
(b)	Compare the salient features of Maxwell-Boltzmann, Bose-Einstein and Fermi-Dirac Distribution Functions.	(5)
(c)	Calculate lattice constant for a rock salt lattice having Molecular Weight 58.45 kg/kmol and density = 2180 kg/m³ .	(5)
(d)	Describe the use of the Zener diode as a voltage regulator.	(5)
(e)	An electron and a proton are moving with the same velocity. Find the ratios of their (i) de-Broglie wavelengths, (ii) phase velocity and (iii) group velocity.	(5)
UNIT - I		
Q2 (a)	State Heisenberg's uncertainty principle. Give experimental proof and hence explain why electrons cannot exist inside the atomic nucleus.	(5)
(b)	The eigen-function for a particle trapped in a 1-D box of width a cm is given by: $\psi(x) = (2/a)^{1/2} \sin(n\pi x/a)$. Find the expectation values of (i) position and (ii) momentum in the ground state.	(5)
OR		
Q3 (a)	Give physical interpretation of wave function ψ and hence explain the condition for orthonormal wave function.	(2)
(b)	Compare the de-Broglie wavelengths for: (i) a 1.0 mg grain of sand blown by the wind at a speed of 20 m/s , and (ii) a 40 keV electron used in a certain electron microscope. What inference will you draw from the above comparison?	(3)
(c)	Define Tunnel Effect and hence prove that the approximate transmission probability for a particle to tunnel through a barrier is: $T = e^{-2k_2 L}$ where $k_2 = [2m(U - E)]^{1/2} / \hbar$, U is the height of the barrier, E is the energy of the particle, L is the width of the barrier and $\hbar$ is reduced Planck's constant.	(5)

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Q. No.	Question	Marks
UNIT - II		
Q4 (a)	What are symmetric and anti-symmetric wave functions? Explain in terms of Bose-Einstein and Fermi-Dirac statistics with suitable examples.	(5)
(b)	Using Maxwell-Boltzmann statistics, derive an expression for the average energy of an ideal gas.	(3)
(c)	Verify that the rms speed of an ideal gas molecule is about 9% greater than its average speed.	(2)
OR		
Q5 (a)	Explain the features of black body spectrum.	(4)
(b)	Describe the features of the Fermi-Dirac distribution function at different temperatures and hence define Fermi level.	(4)
(c)	Write a short note on Dying stars.	(2)
UNIT - III		
Q6 (a)	Draw (110) and (100) planes in a cubic unit cell.	(2)
(b)	Evaluate the packing fraction of BCC and FCC closed-packed structures.	(4)
(c)	Write a short note on the Diamond lattice.	(4)
OR		
Q7 (a)	Derive Bragg's law of crystal diffraction. Discuss the Laue method of crystal structure determination.	(4)
(b)	What are point defects? Differentiate between Schottky and Frenkel defects.	(4)
(c)	The powder of a BCC structured crystal is studied with X-rays of wavelength 2 Å . The (210) reflection is observed at Bragg's angle 35° . Calculate the lattice parameter.	(2)
UNIT - IV		
Q8 (a)	Explain the salient features of the Kronig-Penney model and hence explain the E vs k diagram.	(5)
(b)	Explain the characteristics of a PN junction diode using an energy band diagram.	(5)
OR		
Q9 (a)	Differentiate between intrinsic and extrinsic semiconductors. Show that the Fermi level lies midway between the valence and conduction band for an intrinsic semiconductor.	(7)
(b)	Write a short note on Light Emitting Diode (LED).	(3)

**** End of Question Paper ****